Bihar Engineering University, Patna
End Semester Examination -2023
Semester-IV

Course: B.Tech. ***Time: 03 Hours***

Code: 104401 ***Subject: ECE Analog Circuit*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

(iv) Question No. 1 is compulsory.

Q.1 ***Write the answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which of these are incorrect about Darlington amplifier?

(i) It has a high input resistance (ii) The output resistance is low

(iii) It has a unity voltage gain (iv) It is a current buffer

(b) What is a cascade amplifier?

(i) A cascade of two CE amplifiers (ii) A cascade of two CB amplifiers

(iii) A cascade of CE and CB amplifiers (iv) A cascade of CB and CC amplifiers

(c) What is the output of a class B amplifier for sinusoidal input?

(i) Sinusoidal amplifier (ii) Half-sinusoidal

(iii) Sinusoidal with higher frequency (iv) Square wave

(d) In a class B amplifier, it is found that DC power is 25W, find the ac power.

(i) 10 W (ii) 62.5 W

(iii) 25 W (iv) 50 W

(e) In an oscillator if phase of feedback is same as that of oscillation waveform then feedback is called

(i) Positive feedback (ii) Negative feedback

(iii) Cannot be predicted (iv) Either positive or negative depending upon frequency

(f) For a MOSFET, the pinch-off voltage is -3V. Gate to source voltage is 20V. W/L ratio is 5. Process transconductance parameter is 40μA/V2. Find drain to source current in saturation.

(i) 0.10 mA (ii) 0.05mA

(iii) – 0.05mA (iv) – 50A

(g) For a MOSFET, the pinch-off voltage is -3V. Gate to source voltage is 20V. W/L ratio is 5. Process transconductance parameter is 40μA/V2. Find drain to source current in saturation.

(i) 1100 Ω (ii) 11000 Ω

(iii) 900Ω (iv) 110Ω

(h) The hybrid π – *model is used for*

(i) Analysis of BJT for low frequency

(ii) Analysis of BJT at high frequency

(iii) Analysis of BJT for feedback

(iv) Analysis of BJT for large signal

(i) Which oscillator is characterized by a split capacitor in its tank circuit?

(i) RC phase shift oscillator

(ii) Colpitts oscillator

(iii) Wien bridge oscillator

(iv) none

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(j) The gate voltage in a JFET at which drain current becomes zero is called

(i) pinch-off voltage.
(ii) saturation voltage
(iii) active voltage.
(iv) cut-off voltage.

Q.2 Explain construction and working of Enhancement and Depletion type MOSFET. [14]

Q.3 What is power amplifier. Explain Transformer coupled Class-A amplifier. [14]

Q.4 Barkhausen criteria play an important role while designing Oscillator justify.Explain Hartley and RC phase shift Oscillator. [14]

Q.5 Explain the effect of voltage series and voltage shunt feedback on gain & impedance of amplifier. [14]

Q.6 The BJT is connected as a CE amplifier. Determine Z_i, Z_o, A_v, A_i hybrid model. [14]

Q.7 Describe current mirror circuit in BJT. What are the condition to design current mirror circuit. Obtain the expression for current transfer ratio. [14]

Q.8 Derive the expression for voltage gain and current gain and B.W of multistage amplifier. [14]

Q.9 In brief explain the working of voltage series and current series circuit. Explain the same in case of BJT. [14]

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